

The attenuation constant of the $3x + 1$ cascade: an exact decomposition and a stability program

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Abstract

The full-density program for the $3x + 1$ problem reduces, by the companion papers, to one Open Lemma: the attenuation constant κ of the Krasikov–Lagarias minimum is uniformly bounded away from 1 ($\kappa_\infty = 0.839 \pm 0.002$ measured, nothing proved). This paper is the proof-attempt manuscript for that lemma. The engine is a one-line exact formula: writing the member increments of a sibling triple as $\Delta_i = t + \delta_i$ (common move plus centered fine moves) and the static within-triple gaps as g_i , the increment of the triple minimum is *exactly* $t + S$ with $S = \min_i(\delta_i + g_i)$. The common move therefore passes with slope exactly 1 — recovering the directional low-pass — and *all* attenuation is carried by the scalar S . Squaring gives an exact decomposition, $\kappa^2 = [\text{Var}(t) + 2\text{cov}(t, S) + \text{Var}(S)] / [\text{Var}(t) + \text{Var}(\delta)]$, verified to machine precision on certificate data (0.7901 = 0.7901 at $P = 4$; 0.7431 at $P = 6, k = 13$), so that $\kappa < 1$ if and only if $2\text{cov}(t, S) + \text{Var}(S) < \text{Var}(\delta)$. Both sufficient mechanisms are measured with the right sign and size: $\text{cov}(t, S) < 0$ (selection-weighted mean reversion) and $\text{Var}(S)/\text{Var}(\delta) = 0.639, 0.499 < 1$ (min-variance contraction under intense competition: the selected member is non-argmin with probability 0.658–0.665 $\approx 2/3$, and the median second gap is only $0.13\sigma_\delta$). We prove a clean quantitative min-variance contraction lemma — an exact identity with no hypotheses, plus a contraction bound whose symmetry hypothesis we state explicitly and confront with the data — and assemble the conditional theorem: *if* the mean-reversion sign and the competition condition hold uniformly in depth, *then* $\kappa \leq \kappa_{\max} < 1$ uniformly, the Open Lemma closes, and by the companion chain $\gamma_k \rightarrow 1$: density $x^{1-\varepsilon}$ for every $\varepsilon > 0$. What remains unproved is exactly the two hypotheses; we state the proof obligations, the supporting measurements at every available depth, and the ways the program could fail.

1 Introduction

Let $T(n) = n/2$ for even n , $T(n) = (3n + 1)/2$ for odd n , and let $\pi_1(x)$ count the $n \leq x$ whose T -orbit reaches 1. The difference-inequality method of Krasikov and Lagarias [1] converts a finite feasibility certificate at 3-adic depth k into a density bound $\pi_1(x) \geq x^{\gamma_k}$. The companion papers [4, 5, 6] assemble a program that would push $\gamma_k \rightarrow 1$: the tempering law and the edge-rate theorem $d\gamma/dq = 1/\ln(4/3)$ [4], the mass-conservation, balance, and directional low-pass identities [5], and the synthesis as a single conditional theorem [6]. Every link of that chain is proved except one, the *Open Lemma*: the attenuation constant $\kappa(P; k)$ of the Krasikov–Lagarias minimum — the ratio of the standard deviation of the triple-minimum increment to that of the member increments, under one application of the edge-linearized operator — satisfies

$\kappa(P; k) \leq \kappa_{\max} < 1$ uniformly in scale P and depth k . Measured: $\kappa < 0.95$ everywhere and $\kappa_{\infty} = 0.839 \pm 0.002$, flat across $k = 13, 15, 17, 19$ (Prop. 35 of [7]); proved: nothing.

This paper is the proof attempt. Its contributions are:

1. **The S -formula** (Lemma 2; proved, one line). In the natural coordinates — common move t , centered fine moves δ_i , static gaps g_i — the increment of the triple minimum is exactly $t + S$ with $S = \min_i(\delta_i + g_i)$. The common move passes with slope exactly 1 (recovering the directional low-pass, Prop. 34b of [7], as a corollary), and the entire attenuation mechanism is compressed into the single scalar S .
2. **The exact κ decomposition** (Corollary 5; proved, and verified to machine precision on certificate data):

$$\kappa^2 = [\text{Var}(t) + 2\text{cov}(t, S) + \text{Var}(S)] / [\text{Var}(t) + \text{Var}(\delta)].$$

Hence the Open Lemma is *equivalent* to a two-term inequality: $2\text{cov}(t, S) + \text{Var}(S) < \text{Var}(\delta)$, uniformly.

3. **A min-variance contraction lemma** (Lemma 6; proved). An exact identity — valid with no hypotheses whatsoever — expresses the variance deficit of a minimum as a product of one-sided swap means, $2 \mathbb{E}[W^+] \mathbb{E}[W^-]$, where W is the difference of the competing candidates: the min contracts variance exactly when the *order swaps* with positive probability in both directions. A quantitative contraction bound follows under an explicitly stated reflection-symmetry hypothesis; we say precisely which hypotheses the certificate data supports and which are untested.
4. **The conditional stability theorem** (Theorem 9; proved as stated). If the mean-reversion sign $\text{cov}(t, S) \leq 0$ and the competition condition hold uniformly in depth, then $\kappa \leq \kappa_{\max} < 1$ uniformly: the Open Lemma closes, and by [6] $\gamma_k \rightarrow 1$, i.e. $\pi_1(x) \geq x^{1-\varepsilon}$ for every $\varepsilon > 0$.
5. **The honest gap** (Section 8). What remains is to prove the two hypotheses from the cascade structure. The sign hypothesis is the selection-weighted mean reversion of Prop. 34 [7] (LP duality route sketched); the competition hypothesis ties the gap scale to the increment scale — the ultrametric spacing law (Lemma 26) and the fine-end saturation (Prop. 23: $\text{CV}_1 \rightarrow 0.5136 > 0$). Both are measured across all available depths; neither is proved. We list the falsification risks.

Labels. As in the companions, every assertion carries one of four labels. *Proved*: complete mathematical proof in the text or a cited companion. *Verified*: finite exact or floating-point computation, reproducible from stated parameters. *Measured*: statistical estimate from computed data. *Open*: neither proved nor finitely checkable. Section 9 collects the ledger.

2 Setup and notation

$\beta := \log_2 3$. For $j \geq 1$, $[3^j] := \{m \bmod 3^j : m \equiv 2 \pmod{3}\}$. Fix $k \geq 2$ and $\lambda \in (1, 2]$. The nontrivial-tree Krasikov–Lagarias system $L_k(\lambda)$ asks for $c : [3^k] \rightarrow \mathbb{R}_{>0}$ with, for every m ,

$$m \equiv 2 \pmod{9} : \quad c(m) \leq \lambda^{-2} c(4m) + \lambda^{\beta-2} \bar{c}\left(\frac{4m-2}{3}\right), \quad (1)$$

$$m \equiv 5 \pmod{9} : \quad c(m) \leq \lambda^{-2} c(4m), \quad (2)$$

$$m \equiv 8 \pmod{9} : \quad c(m) \leq \lambda^{-2} c(4m) + \lambda^{\beta-1} \bar{c}\left(\frac{2m-1}{3}\right), \quad (3)$$

where $4m$ is reduced mod 3^k , branch targets mod 3^{k-1} , and $\bar{c}(t) := \min\{c(t), c(t + 3^{k-1}), c(t + 2 \cdot 3^{k-1})\}$ is the minimum over the three refinements of t — the *sibling triple*. Feasibility implies $\pi_1(x) \geq x^\gamma$ with $\gamma = \log_2 \lambda$ [1]. The certificate data below refer to the critical (Perron) certificates `cert_k13` etc. of [4]; P denotes the 3-adic scale of the difference being tracked, as in [5].

2.1 Triple coordinates

Consider a sibling triple with member levels $x = (x_1, x_2, x_3)$ — the logarithmic certificate values whose minimum the system reads — and let one application of the edge-linearized operator move member i by the increment Δ_i . Decompose

$$\Delta_i = t + \delta_i, \quad t := \frac{1}{3} \sum_i \Delta_i, \quad \sum_i \delta_i = 0, \quad (4)$$

so t is the *common (coarse) move* of the triple and δ_i the centered *fine moves*. Let

$$g_i := x_i - \min_j x_j \geq 0 \quad (5)$$

be the static within-triple *gaps* (at least one $g_i = 0$), and write $g_{(1)} \leq g_{(2)} \leq g_{(3)}$ for their order statistics ($g_{(1)} = 0$; $g_{(2)}$ is the *second gap*, the margin by which the runner-up trails the argmin).

Statistics are taken over the population of sibling triples of a certificate at a given scale P : $\text{Var}(t)$ and $\text{cov}(t, \cdot)$ are over triples; $\text{Var}(\delta) := \mathbb{E}[\delta_J^2]$ is the pooled fine variance, J a uniformly random member label independent of the triple (note $\mathbb{E}[\delta_J] = 0$ and $\text{cov}(t, \delta_J) = \mathbb{E}[t \mathbb{E}(\delta_J \mid \text{triple})] = 0$, both exact); $\sigma_\delta := \text{Var}(\delta)^{1/2}$.

Definition 1 (Attenuation constant; Def. 10 of [5]). $\kappa(P; k) := \frac{\text{std}(\min_i(x_i + \Delta_i) - \min_i x_i)}{\text{std}(\Delta_J)}$, the ratio of the standard deviation of the triple-minimum increment to that of the pooled member increments.

3 The S -formula

Lemma 2 (S -formula). (Proved.) *For any triple, with the coordinates (4)–(5),*

$$\min_i(x_i + \Delta_i) - \min_i x_i = t + S, \quad S := \min_i(\delta_i + g_i). \quad (6)$$

Proof. $\min_i(x_i + t + \delta_i) = t + \min_i(\min_j x_j + g_i + \delta_i) = \min_j x_j + t + \min_i(g_i + \delta_i)$. $\square$

Verified: identity (6) holds to 1.8×10^{-15} (maximum absolute residual, i.e. floating-point roundoff) on 150,000 certificate triples.

Corollary 3 (Slope exactly one; all attenuation in S). (Proved.) *The triple minimum passes the common move with slope exactly 1: if $\delta = 0$ then $S = \min_i g_i = 0$ and the min-increment equals t . This recovers the directional low-pass, Prop. 34b of [7] ($\min(x + c\mathbf{1}) = \min(x) + c$), as the $\delta = 0$ case of Lemma 2. In general the min-increment deviates from the common move by exactly the scalar S , which depends only on the fine moves and the gaps; moreover*

$$\min_i \delta_i \leq S \leq \delta_{i_0}, \quad i_0 := \operatorname{argmin}_i x_i,$$

the upper bound being the old-argmin half of the sandwich inequality of Prop. 33 [7].

Proof. $S \leq \delta_{i_0} + g_{i_0} = \delta_{i_0}$ since $g_{i_0} = 0$; $S \geq \min_i \delta_i$ since $g_i \geq 0$. $\square$

Remark 4 (Relation to the clipping decomposition). Prop. 33 of [7] decomposes the min-increment as $\bar{D} + R$, member mean plus min-correction, and finds the attenuation concentrated in the covariance $\operatorname{cov}(\bar{D}, R) < 0$. The coordinates (t, S) of Lemma 2 are the same budget in a sharper basis: $t = \bar{D}$ and $S = R$, but now S has a *closed form* — a minimum of fine moves shifted by gaps — rather than being defined as a residual. Every structural question about the attenuation becomes a question about the scalar random variable $S = \min_i (\delta_i + g_i)$.

4 The exact decomposition of κ

Corollary 5 (Exact κ decomposition). (Proved; verified exactly on certificate data.)

$$\kappa^2 = \frac{\operatorname{Var}(t) + 2\operatorname{cov}(t, S) + \operatorname{Var}(S)}{\operatorname{Var}(t) + \operatorname{Var}(\delta)}, \quad (7)$$

and consequently

$$\kappa < 1 \iff 2\operatorname{cov}(t, S) + \operatorname{Var}(S) < \operatorname{Var}(\delta). \quad (8)$$

Proof. Numerator: by Lemma 2 the min-increment is $t + S$, and $\operatorname{Var}(t + S) = \operatorname{Var}(t) + 2\operatorname{cov}(t, S) + \operatorname{Var}(S)$. Denominator: with J a uniform member label independent of the triple, $\operatorname{Var}(\Delta_J) = \operatorname{Var}(t + \delta_J) = \operatorname{Var}(t) + \operatorname{Var}(\delta)$, the cross term vanishing exactly because $\mathbb{E}[\delta_J \mid \text{triple}] = 0$. Divide. The equivalence (8) is numerator minus denominator. $\square$

Verified: on `cert_k13`, the right side of (7) equals the directly computed κ^2 to all displayed digits: 0.7901 = 0.7901 at $P = 4$ and 0.7431 = 0.7431 at $P = 6$ (i.e. $\kappa = 0.889, 0.862$, matching the independently measured attenuation series $0.908 \rightarrow 0.841$ of [5] at those scales).

The decomposition splits the Open Lemma into exactly two mechanisms:

- (M1) *Mean reversion:* $\operatorname{cov}(t, S) \leq 0$ — the selected fine move anti-correlates with the common move;
- (M2) *Min-variance contraction:* $\operatorname{Var}(S) < \operatorname{Var}(\delta)$ — the minimum over competing candidates is less variable than a single member's fine move.

Either mechanism alone, if quantitatively uniform, would suffice; the data (next section) shows *both* present at every measured scale and depth.

5 Measured inputs

All quantities on `cert_k13`; covariances in normalized units $V := \text{Var}(t) + \text{Var}(\delta) = 1$.

quantity	$P = 4$	$P = 6$
κ^2 (both sides of (7))	0.7901	0.7431
$\text{Var}(S)/\text{Var}(\delta)$	0.639	0.499
$\text{cov}(t, S)$	-0.0799	-0.0517
$\text{Pr}(S \text{ selects a non-argmin member})$	0.658	0.665
median $g_{(2)}/\sigma_\delta$	0.130	0.127
implied $\text{Var}(\delta)/V$ (from (7))	0.139	0.306

Reading of the table (all four rows are *measured*; the last row is arithmetic from the rows above and closes the budget):

- *Both mechanisms are active with the right sign.* $\text{cov}(t, S) < 0$ at both scales (M1), and $\text{Var}(S)/\text{Var}(\delta) < 1$ at both scales (M2). The attenuation budget $1 - \kappa^2 = -2\text{cov}(t, S) + (1 - \frac{\text{Var}(S)}{\text{Var}(\delta)}) \frac{\text{Var}(\delta)}{V}$ splits as $0.1598 + 0.0501$ at $P = 4$ (covariance-dominated) and $0.1034 + 0.1535$ at $P = 6$ (contraction-dominated): the two mechanisms trade weight with scale while their sum grows.
- *The competition is intense.* The median second gap is only $0.13 \sigma_\delta$: the runner-up typically trails the argmin by an eighth of a fine-move standard deviation, so a single application routinely reorders the triple. Consistently, S selects a member other than the static argmin with probability $0.658\text{--}0.665$ — within one percent of the value $2/3$ that full member-exchangeability with gap-blind selection would predict. This is the data's vote for the exchangeability hypothesis of Lemma 6(d) below.
- *The fine fraction grows with scale.* $\text{Var}(\delta)/V$ rises from 0.14 to 0.31 between $P = 4$ and $P = 6$: deep scales are fine-dominated, which is where mechanism (M2) has the most leverage — consistent with the deepening of the measured attenuation ($\kappa = 0.908 \rightarrow 0.841$ across P , [5]).

6 The min-variance contraction

Mechanism (M2) asks when a minimum of competing random candidates is *less* variable than a single candidate. This is not automatic — a minimum of comonotone candidates just tracks the lowest one, with no contraction — so the content must lie in *competition*. The following lemma isolates exactly that: an identity with no hypotheses whose deficit term is nonzero precisely when the *order swaps*, plus a quantitative bound under hypotheses we state explicitly.

For a real random variable W write $W^+ = \max(W, 0)$, $W^- = \max(-W, 0)$.

Lemma 6 (Min-variance contraction). (Proved; hypotheses as stated per part.) *Let A, B be square-integrable random variables on a common probability space, $W := A - B$, $m := \min(A, B)$, $M := \max(A, B)$, and $\sigma^2 := \max(\text{Var} A, \text{Var} B)$.*

(a) (Exact identity; no hypotheses.)

$$\text{Var}(m) + \text{Var}(M) = \text{Var}(A) + \text{Var}(B) - 2\mathbb{E}[W^+] \mathbb{E}[W^-]. \quad (9)$$

The deficit $2\mathbb{E}[W^+]\mathbb{E}[W^-]$ is strictly positive if and only if $\Pr(A > B) > 0$ and $\Pr(B > A) > 0$: variance is destroyed exactly when the order swaps.

(b) (Splitting min from max.) With $\Delta_{\text{skew}} := \text{Var}(m) - \text{Var}(M)$,

$$\text{Var}(m) = \frac{1}{2}(\text{Var}A + \text{Var}B) - \mathbb{E}[W^+]\mathbb{E}[W^-] + \frac{1}{2}\Delta_{\text{skew}}.$$

If the law of (A, B) is invariant under the anti-diagonal reflection $(A, B) \mapsto (s - B, s - A)$ for some constant s , then $\Delta_{\text{skew}} = 0$ and

$$\text{Var}(\min(A, B)) = \frac{1}{2}(\text{Var}A + \text{Var}B) - \mathbb{E}[W^+]\mathbb{E}[W^-] \leq \sigma^2 - \mathbb{E}[W^+]\mathbb{E}[W^-].$$

(c) (Quantitative competition bound.) If, in addition to the reflection hypothesis of (b), there are $p_0 > 0$, $c > 0$ with

$$\Pr(W \geq c\sigma) \geq p_0 \quad \text{and} \quad \Pr(-W \geq c\sigma) \geq p_0$$

(two-sided order-swap frequency), then

$$\text{Var}(\min(A, B)) \leq (1 - \varepsilon(p_0, c))\sigma^2, \quad \varepsilon(p_0, c) := p_0^2 c^2 > 0.$$

(d) (Exchangeable triple.) Let (Y_1, Y_2, Y_3) be exchangeable with common variance σ^2 and mean μ , order statistics $Y_{(1)} \leq Y_{(2)} \leq Y_{(3)}$, and set $a_i := \mathbb{E}[Y_{(i)}] - \mu$. Then, with no further hypotheses,

$$\sum_i \text{Var}(Y_{(i)}) = 3\sigma^2 - \sum_i a_i^2,$$

and if additionally the vector is centrally symmetric ($2\mu\mathbf{1} - Y \stackrel{d}{=} Y$), then $\text{Var}(Y_{(1)}) = \text{Var}(Y_{(3)})$, $a_2 = 0$, and

$$\text{Var}(\min_i Y_i) = \frac{1}{2}\left(3\sigma^2 - \frac{1}{2}(\mathbb{E}[Y_{(3)} - Y_{(1)}])^2 - \text{Var}(Y_{(2)})\right).$$

In particular $\text{Var}(\min) < \sigma^2$ whenever $(\mathbb{E}[\text{range}])^2 + 2\text{Var}(\text{median}) > 2\sigma^2$.

Proof. (a) Pointwise $\{m, M\} = \{A, B\}$, so $\mathbb{E}[m^2 + M^2] = \mathbb{E}[A^2 + B^2]$. From $m = \frac{1}{2}(A+B) - \frac{1}{2}|W|$ and $M = \frac{1}{2}(A+B) + \frac{1}{2}|W|$, with $\mu := \frac{1}{2}(\mathbb{E}A + \mathbb{E}B)$, $d := \mathbb{E}|W|$, $b := \mathbb{E}W$:

$$(\mathbb{E}m)^2 + (\mathbb{E}M)^2 = 2\mu^2 + \frac{d^2}{2}, \quad (\mathbb{E}A)^2 + (\mathbb{E}B)^2 = 2\mu^2 + \frac{b^2}{2}.$$

Subtracting, $\text{Var}m + \text{Var}M = \text{Var}A + \text{Var}B - \frac{1}{2}(d^2 - b^2)$. Since $|W| = W^+ + W^-$ and $W = W^+ - W^-$, $d - b = 2\mathbb{E}[W^-]$ and $d + b = 2\mathbb{E}[W^+]$, so $d^2 - b^2 = 4\mathbb{E}[W^+]\mathbb{E}[W^-]$, giving (9). The deficit is positive iff both factors are, i.e. iff the order swaps both ways with positive probability.

(b) The first display is (9) rearranged. Under the reflection, $\min(s-B, s-A) = s - \max(A, B)$, so $m \stackrel{d}{=} s - M$ and $\text{Var}m = \text{Var}M$.

(c) $\mathbb{E}[W^+] \geq c\sigma \Pr(W \geq c\sigma) \geq p_0 c\sigma$ and symmetrically for W^- ; insert into (b) and use $\frac{1}{2}(\text{Var}A + \text{Var}B) \leq \sigma^2$.

(d) Pointwise $\sum_i Y_{(i)}^2 = \sum_i Y_i^2$, so the second moments agree; exchangeability gives $\mathbb{E}Y_i = \mu$ for all i , and $\sum_i a_i = 0$, so $\sum_i (\mathbb{E}Y_{(i)})^2 = 3\mu^2 + \sum_i a_i^2$, proving the first display. Central symmetry maps $Y_{(1)} \mapsto 2\mu - Y_{(3)}$ and $Y_{(2)} \mapsto 2\mu - Y_{(2)}$, so $\text{Var}Y_{(1)} = \text{Var}Y_{(3)}$, $a_2 = 0$, and $a_3 = -a_1 = \frac{1}{2}\mathbb{E}[Y_{(3)} - Y_{(1)}]$, whence $\sum a_i^2 = \frac{1}{2}(\mathbb{E}[\text{range}])^2$; solve the first display for $\text{Var}(Y_{(1)})$. $\square$

Remark 7 (Near-ties plus dispersion imply order swaps). The intuitive competition event — “at least two candidates lie within $c_0\sigma_\delta$ of each other” — does not by itself produce contraction: two comonotone candidates a hair apart never swap, and part (a) then gives deficit zero. What near-ties buy is *leverage* for the fine-move dispersion. Concretely, for the two lowest-gap members a ($\arg\min$, $g_a = 0$) and b (runner-up, $g_b = g_{(2)}$), $W = Y_a - Y_b = (\delta_a - \delta_b) - g_{(2)}$; on the joint event $\{g_{(2)} \leq c_0\sigma_\delta\} \cap \{\delta_a - \delta_b \geq c_1\sigma_\delta\}$ with $c_1 > c_0$ one has $W \geq (c_1 - c_0)\sigma_\delta$, while $\{-W \geq c_1\sigma_\delta\} \supseteq \{\delta_b - \delta_a \geq c_1\sigma_\delta\} \cap \{g_{(2)} \geq 0\}$ costs nothing. So: *small static gaps* (measured: median $g_{(2)} = 0.13\sigma_\delta$) *plus two-sided anti-concentration of the fine-move differences* (non-comonotonicity) deliver the two-sided swap hypothesis of part (c) with $c = c_1 - c_0$. The near-tie condition and the swap condition are the static and dynamic faces of the same competition.

Remark 8 (Which hypotheses does the data support?). Honesty requires separating the three hypotheses used above.

- *Exchangeability* (part (d), and implicitly the “no distinguished member” reading of (c)): supported. Under member-exchangeability with gaps small relative to fine moves, the selected member is nearly uniform over the triple, predicting $\Pr(\text{non-argmin selection}) = 2/3 = 0.667$; measured 0.658 and 0.665 (Section 5).
- *Competition / order swaps*: supported. Median $g_{(2)}/\sigma_\delta = 0.130, 0.127$ — the near-tie regime of Remark 7 — and the 0.66 non-argmin selection rate is direct evidence that the order swaps at nearly the exchangeable ceiling.
- *Reflection / central symmetry* (the hypothesis that kills the skew term Δ_{skew}): *untested*. We have not separately measured $\text{Var}(\min) - \text{Var}(\max)$ on the certificate triples; a persistent positive skew would weaken part (c)’s bound by $\frac{1}{2}\Delta_{\text{skew}}$. What *is* directly verified is the conclusion itself: $\text{Var}(S)/\text{Var}(\delta) = 0.639, 0.499 < 1$ — so the package (swap frequency + approximate symmetry) is consistent ex post, but the symmetry hypothesis is a genuine assumption of the clean statement, not a theorem about the cascade. We flag it as such in Theorem 9 and Section 8.

Note also a bookkeeping point: parts (a)–(d) control $\text{Var}(\min Y)$ relative to $\text{Var}(Y_i) = \text{Var}(\delta_i + g_i)$, which exceeds $\text{Var}(\delta)$ when the gaps fluctuate across triples. Converting the contraction into the form needed by (8), $\text{Var}(S) \leq (1 - \varepsilon_0)\text{Var}(\delta)$, therefore consumes part of the deficit to pay for the gap variance — which is affordable precisely in the near-tie regime where gaps are small on the σ_δ scale. This is where the gap-to-increment scale ratio enters irreducibly; see Section 8.

7 The conditional stability theorem

Theorem 9 (Conditional uniform attenuation). (Proved as stated; the hypotheses are the open content.) *Suppose there exist constants $\varepsilon_0 > 0$ and $\varphi_0 > 0$ such that, for every scale P and every depth k , in the stationary (Perron) state of the critical Krasikov–Lagarias system:*

(i) (Mean-reversion sign) $\text{cov}(t, S) \leq 0$;

(ii) (Uniform competition) $\text{Var}(S) \leq (1 - \varepsilon_0)\text{Var}(\delta)$ and $\text{Var}(\delta) \geq \varphi_0(\text{Var}(t) + \text{Var}(\delta))$.

Then

$$\kappa(P; k)^2 \leq 1 - \varepsilon_0 \varphi_0 =: \kappa_{\max}^2 < 1 \quad \text{for all } P, k :$$

the Open Lemma of [5, 6] holds with $\kappa_{\max} = (1 - \varepsilon_0 \varphi_0)^{1/2}$, and consequently, by the conditional theorem of [6] (whose remaining five links are proved there), $\gamma_k \rightarrow 1$ and

$$\pi_1(x) \geq x^{1-\varepsilon} \quad \text{for every } \varepsilon > 0.$$

Proof. By Corollary 5 and (i), then (ii),

$$\kappa^2 \leq \frac{\text{Var}(t) + \text{Var}(S)}{\text{Var}(t) + \text{Var}(\delta)} \leq \frac{\text{Var}(t) + (1 - \varepsilon_0)\text{Var}(\delta)}{\text{Var}(t) + \text{Var}(\delta)} = 1 - \varepsilon_0 \frac{\text{Var}(\delta)}{\text{Var}(t) + \text{Var}(\delta)} \leq 1 - \varepsilon_0 \varphi_0.$$

□

Remark 10 (Measured values of the constants). The hypotheses hold with room at every measured point: $\text{cov}(t, S) = -0.0799$, $-0.0517 < 0$; $\varepsilon_0 = 1 - 0.639 = 0.361$ at the weakest measured scale; $\varphi_0 = 0.139$ at the coarsest measured scale (and growing with P). These give $\kappa_{\max} = (1 - 0.361 \times 0.139)^{1/2} = 0.975$ — a valid but crude bound, since the proof spends hypothesis (i) only on its sign. Keeping the measured covariance yields the measured $\kappa = 0.889$ at $P = 4$; the theorem’s job is uniformity, not sharpness. Note also that Lemma 6(c), under its stated hypotheses with (p_0, c) uniform in depth and the gap-variance bookkeeping of Remark 8, is precisely a mechanism that would deliver hypothesis (ii); and Prop. 34’s binding rigidity [7] is the mechanism proposed for (i). Neither delivery is yet a proof.

8 The honest gap

Theorem 9 is an exact reduction, not a resolution: the Open Lemma is now equivalent (via the verified identity (7)) to proving hypotheses (i) and (ii) *from the cascade structure*. We record what each requires, what is measured, and what would falsify the program.

8.1 Proving (i): selection-weighted mean reversion

$\text{cov}(t, S) \leq 0$ says: when the triple’s common move is up, the selected member’s fine move is preferentially down, and vice versa. This is exactly the phenomenon isolated in Prop. 34 of [7]: the antisymmetric min-correction (± 0.026) is carried equally by switch and no-switch events, so the mechanism is not clipping-at-switches but *binding-constraint rigidity* — the argmin member is the binding entry of the rows that read it, and binding entries, pinned by the equation network through complementary slackness, co-move less than the triple average. The proof route (sketched in [5], Sect. “The LP-duality route”) is:

1. at the critical λ_k the certificate is an LP optimum, and each triple’s argmin carries a strictly positive dual weight;
2. show the dual weights concentrate on binding entries with pinning strength bounded below uniformly, using the mass-conservation identity (normalizing total dual mass) and the balance identity (preventing dual mass from draining across the scale hierarchy), both proved in [5];

3. sensitivity analysis then bounds the binding member’s response slope below the slack members’, giving $\text{cov}(t, S) \leq 0$ in the stationary state.

Step 2 is the missing arithmetic input; nothing here is circular with the present paper, but nothing here is proved either.

8.2 Proving (ii): the gap scale versus the increment scale

Uniform competition requires that the static gaps do not outgrow the fine-move dispersion: the near-tie ratio $g_{(2)}/\sigma_\delta$ must stay bounded (measured: median 0.130, 0.127 at $P = 4, 6$) and the fine fraction $\text{Var}(\delta)/V$ must stay bounded below (measured: 0.139, 0.306 and growing with scale). Two measured laws of [7] are the natural sources:

- *The ultrametric spacing law* (Lemma 26): the triple-CV of the certificate at class-spacing s depends to leading order only on $v_3(s)$, with an approximate two-term closure. This is exactly a statement tying the gap distribution at each scale to the certificate’s own CV profile — i.e. tying the gap scale to the increment scale. Measured, with an emergent-coefficient caveat; not proved.
- *Fine-end saturation* (Prop. 23): $\text{CV}_1(k) = 0.5136 - 0.337(0.910)^k$ to residual 10^{-6} across nine depths — the intra-triple variation converges to a *nonzero* limit. This is what keeps σ_δ (and with it the fine fraction φ_0) alive as $k \rightarrow \infty$; its failure ($\text{CV}_1 \rightarrow 0$) would kill competition and push $\kappa \rightarrow 1$. Measured with striking precision; not proved.

8.3 What is measured across depth

The program’s empirical safety margin is Prop. 35 of [7]: $\kappa < 0.95$ at every scale and depth, with the deep-scale value flat, $\kappa_{\text{deep}} = 0.841, 0.840, 0.837, 0.838$ at $k = 13, 15, 17, 19$, i.e. $\kappa_\infty = 0.839 \pm 0.002$. And by the scalar closure (Prop. 47 of [7]), the variance budget of the exact transport identity closes to 0.06% with $\kappa(P)$ as the only nonlinear unknown: the entire cascade is governed by a one-dimensional map whose measured trajectory descends ($0.917 \rightarrow 0.899$ across P , deep limit 0.839). The stability program of the title is precisely: prove that this one-dimensional fixed point sits below 1 — which Theorem 9 reduces to the sign (i) and the competition (ii).

8.4 Falsification risks

Stated symmetrically, as in the companions:

1. *Sign creep.* $\text{cov}(t, S)$ could drift to 0 or turn positive at depths beyond measurement. The measured normalized value shrinks from -0.0799 ($P=4$) to -0.0517 ($P=6$); on the present data the contraction term more than compensates (total attenuation grows), but a proof of (i) must rule out asymptotic sign loss, and the trend is a genuine caution.
2. *Competition dilution.* The gap scale could grow relative to σ_δ at deep scales or depths ($g_{(2)}/\sigma_\delta$ measured flat at 0.13, but only to $P = 6$, $k = 19$), or the fine fraction could collapse ($\varphi_0 \rightarrow 0$) if Prop. 23’s saturation is preasymptotic.
3. *Symmetry breakdown.* The clean contraction bound assumes reflection symmetry; a persistent skew $\Delta_{\text{skew}} > 0$ would eat into the deficit $\mathbb{E}[W^+]\mathbb{E}[W^-]$. This hypothesis is currently

untested on certificate data (Remark 8) and is the cheapest of the risks to retire: measure $\text{Var}(\min) - \text{Var}(\max)$ across P and k .

4. *Exchangeability breakdown.* The non-argmin selection rate could drift from $2/3$ at large k , signaling a distinguished member and invalidating the symmetric reading of the competition bound.
5. *Preasymptotic flatness.* The depth-flatness of κ_{deep} over $k = 13\text{--}19$ spans a factor 3^6 in modulus but is still a finite window; all conclusions drawn from it are measured, not proved, and are labeled so.

9 Status ledger

Proved	S -formula (6) and slope-one corollary (Lemma 2, Cor. 3), recovering Prop. 34b; exact decomposition (7) and criterion (8) (Cor. 5); min-variance identities and contraction bounds under stated hypotheses (Lemma 6(a)–(d)); conditional uniform attenuation (Thm. 9); the five other links of the $\gamma \rightarrow 1$ chain [4, 5, 6].
Verified	S -formula to 1.8×10^{-15} on 150,000 certificate triples; decomposition closure $0.7901 = 0.7901$ ($P=4$), 0.7431 ($P=6$), $k = 13$.
Measured	$\text{Var}(S)/\text{Var}(\delta) = 0.639, 0.499$; $\text{cov}(t, S) = -0.0799, -0.0517$; non-argmin selection $0.658, 0.665$ (vs. $2/3$ exchangeable); median $g_{(2)}/\sigma_\delta = 0.130, 0.127$; implied fine fraction $0.139, 0.306$; κ flat at 0.839 ± 0.002 over $k = 13\text{--}19$ (Prop. 35); $\text{CV}_1 \rightarrow 0.5136 > 0$ (Prop. 23); spacing law (Lemma 26); scalar closure to 0.06% (Prop. 47) [7].
Open	Hypothesis (i): $\text{cov}(t, S) \leq 0$ uniformly, from binding-constraint rigidity (Prop. 34; LP-duality route, step 2 missing). Hypothesis (ii): uniform competition — gap scale tied to increment scale (Lemma 26) and fine-end saturation (Prop. 23), neither proved. Reflection-symmetry hypothesis of Lemma 6(b)–(c): untested on certificate data.

Acknowledgment of method

The computations, fits, and drafts were produced in an extended human–AI collaboration; every proved statement above has a self-contained proof in the text or in a cited companion, independent of that provenance.

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